VQ-MAR Phase 1: Georgia Aquifer

SSURGO Data Documentation

Companion Reference for georgia/raw/ssurgo/ and scripts/ssurgo_scripts/

Version 2.1 — May 2026

1. Overview

This document describes the SSURGO (Soil Survey Geographic Database)[2] data files compiled for VQ-MAR Phase 1[1], the Georgia aquifer pilot study. The data was queried programmatically from the USDA NRCS Soil Data Access (SDA) web service[3] and covers the Dougherty Plain region of southwest Georgia[5] — a karst province of the Upper Floridan Aquifer system spanning 10 counties (Sumter, Lee, Terrell, Webster, Schley, Macon, Dooly, Crisp, Worth, and Dougherty).

The study area bounding box is 31.60°N–32.50°N latitude by 84.60°W–83.60°W longitude, approximately 100 km north-south by 95 km east-west. Within this area, candidate recharge sites were placed on regular grids at two problem sizes (20 and 50 sites) to support the VQ-MAR scaling study described in the IEEE QCE 2026 paper.

The SSURGO data provides the foundational soil hydraulic properties that determine whether water can physically infiltrate from the surface into the aquifer. In the VQ-MAR QUBO formulation, SSURGO contributes 30% of the linear site-benefit coefficient w_i (the single largest weight among the five data sources) and also provides hard exclusion flags e_i for sites with unsuitable depth or soil conditions.

Data source: USDA NRCS Soil Data Access (SDA) web service, https://sdmdataaccess.nrcs.usda.gov

Database: SSURGO (Soil Survey Geographic Database), the most detailed soil survey product published by the USDA Natural Resources Conservation Service.

Access method: HTTP POST queries against the SDA Tabular REST endpoint. Spatial assignment used the SDA_Get_Mukey_from_intersection_with_WktWgs84 stored procedure[4] for point-in-polygon lookup.

Cost: $0. All data is publicly funded and freely available.

Date collected: March 2026.

Data location: vqmar-georgia01/georgia/raw/ssurgo/

Scripts location: vqmar-georgia01/scripts/ssurgo_scripts/ — contains fetch_ssurgo.py, fetch_ssurgo_spatial.py, and transform_ssurgo.py (the scoring and exclusion-flag pipeline).

2. How the Data Is Applied

2.1 Role in the QUBO Formulation

The VQ-MAR paper[1] formulates MAR site selection[6] as a Quadratic Unconstrained Binary Optimization (QUBO) problem. The Hamiltonian to be minimized is:

$$H(\mathbf{x}) = -\Sigma\, w_i \cdot x_i + \lambda(\Sigma\, C_i \cdot x_i - B)^2 + \mu\Sigma\, e_i \cdot x_i + \Sigma\, M_{ij} \cdot x_i \cdot x_j$$

SSURGO data feeds three of these terms:

Linear benefit w_i (30% weight): The infiltration score derived from saturated hydraulic conductivity (ksat_r), hydrologic soil group (hydgrp)[8], and soil texture class (texcl) determines how much water each candidate site can absorb. This is the largest single contributor to w_i. Per Equation (3) of the paper:

$$w_i = \alpha R_i + \beta S_i + \gamma L_i$$

where R_i (recharge potential) and S_i (subsurface suitability) are both driven primarily by SSURGO properties.

Exclusion flags e_i: Sites where the minimum annual depth to water table (wtdepannmin) is less than 300 cm, or where soil drainage class indicates permanent saturation, receive $e_i = 1$ and are either removed from the candidate set or penalized with coefficient μ in the QUBO. This prevents the optimizer from selecting sites where there is physically no room in the vadose zone to store recharged water.

Infrastructure cost C_i: Depth to water table inversely correlates with the cost of constructing surface infiltration basins. Shallow water tables increase excavation complexity and reduce the effective storage volume. The cost term C_i in the budget constraint includes a SSURGO-derived depth component per Equation (15) of the paper:

$$C_i = c_0 + c_1 \cdot \tilde{D}_i + c_2 \cdot (1 - L_i) + c_3 \cdot \tilde{A}_i$$

2.2 Data Flow in the VQ-MAR Pipeline

The SSURGO data flows through the pipeline as follows:

1. Candidate site grids (candidate_grid_{20,50}.csv) define geographic locations.

2. Site-to-mukey mapping (site_mukeys_{20,50}.csv) assigns each site to its SSURGO map unit.

3. Tabular soil properties (ssurgo_soil_properties.csv, ssurgo_texture.csv, ssurgo_water_table.csv) are joined to each site via mukey.

4. Per-site component scores are aggregated to a single infiltration score (weighted by comppct_r across soil horizons).

5. The infiltration score, exclusion flags, and cost modifiers are passed to the Qiskit QuadraticProgram[7] as linear coefficients and constraints.

2.3 Integration with Other Data Sources

SSURGO provides the subsurface suitability layer. It is combined with four other data sources to compute the full site benefit w_i: USGS NWIS groundwater levels[9] (25% weight, storage capacity), USGS NLCD land cover[10] (15%, surface availability), NOAA NCEI climate data[11] (15%, recharge timing), and OpenStreetMap infrastructure[12] (15%, access cost). The QUBO is then solved using classical baselines (greedy, simulated annealing, genetic algorithm) and a quantum-compatible QAOA pathway[13] on IBM Quantum via Qiskit Runtime.

3. Data Definition

3.1 Key Terms

Term	Definition
mukey	Map Unit Key. The unique identifier for a SSURGO soil map unit — a delineated area on the landscape with a defined set of soil properties. Each candidate site falls within exactly one map unit.
cokey	Component Key. A map unit contains one or more soil components (e.g., 60% Tifton, 25% Fuquay, 15% Pelham). Each component has its own properties. The cokey uniquely identifies a component within a map unit.
chkey	Horizon Key. Each soil component is divided into vertical layers (horizons) with different properties. The chkey uniquely identifies one horizon within a component.
comppct_r	Component Percentage (representative). The percentage of the map unit area that this component occupies. Used as a weighting factor when aggregating horizon-level properties to the site level.
QUBO	Quadratic Unconstrained Binary Optimization. The mathematical formulation used by VQ-MAR to encode the MAR site selection problem for quantum-compatible solvers.
w_i	Linear site-benefit coefficient in the QUBO. Higher w_i means the site is more suitable for MAR. SSURGO contributes 30% of this value.
e_i	Exclusion flag. Binary indicator (0 or 1) marking sites that are physically unsuitable for MAR. Penalized in the QUBO with coefficient μ.
MAR	Managed Aquifer Recharge. The deliberate replenishment of groundwater through engineered infiltration basins, injection wells, or other infrastructure.
Ksat	Saturated Hydraulic Conductivity. The rate at which water flows through fully saturated soil, in µm/sec. The single most important soil property for MAR site viability.
SDA	Soil Data Access. The USDA NRCS web service providing programmatic access to the SSURGO database.

3.2 File Name Conventions

Pattern	Meaning
candidate_grid_{N}	Grid of N candidate site locations (20 or 50) distributed across the study area. These represent the binary decision variables x_i in the QUBO.
site_mukeys_{N}	Mapping from each of the N candidate sites to its SSURGO map unit key (mukey). The join key connecting site locations to soil properties.
ssurgo_soil_properties	Component-level and horizon-level soil physical properties (Ksat, AWC, sand/clay content, bulk density, organic matter) for all mukeys in the candidate grids.
ssurgo_texture	Soil texture class names (sand, loamy sand, sandy clay loam, etc.) from the chtexturegrp/chtexture tables. Separate file because texture lives in a different SSURGO table than the other horizon properties.
ssurgo_water_table	Aggregate map-unit-level water table and bedrock depth from the muaggatt table. One row per mukey (not per horizon).

3.3 Column Definitions — ssurgo_soil_properties.csv

Column	Units	Source Table	Definition and QUBO Role
cokey	—	component	Component key. Foreign key linking to component-level metadata. One map unit may have multiple components.
mukey	—	component	Map unit key. The primary join column to site_mukeys_{N}.csv. All soil properties for a site are accessed through this key.
comppct_r	%	component	Representative component percentage (15–100%). Used to weight-average properties when aggregating multiple components to a single site score. Only components with comppct_r ≥ 15% are included.
compname	—	component	Soil series name (e.g., Tifton, Orangeburg, Faceville). Identifies the soil type. The dataset contains 37 unique series across the study area.
hydgrp	A/B/C/D	component	Hydrologic Soil Group. A = high infiltration (sand/gravel), B = moderate, C = low, D = very low (clay). Dual groups (A/D, B/D, C/D) indicate drained/undrained states. Scored for QUBO: A=1.0, B=0.7, C=0.4, D=0.1.
drainagecl	—	component	Natural drainage class. Seven values from 'Excessively drained' to 'Very poorly drained'. Used for exclusion: 'Very poorly drained' and 'Poorly drained' sites may receive $e_i = 1$ if water table is also shallow.
slope_r	%	component	Representative slope gradient (1–30%). Steeper slopes increase surface runoff at the expense of infiltration. Used as a penalty modifier on infiltration score.
chkey	—	chorizon	Horizon key. Links to the specific soil layer. Multiple horizons per component represent the vertical soil profile.
hzdept_r	cm	chorizon	Horizon top depth. Depth from the soil surface to the top of this horizon. Used to identify the surface horizon (hzdept_r = 0) which controls initial infiltration.
hzdepb_r	cm	chorizon	Horizon bottom depth. Depth from the soil surface to the bottom of this horizon. hzdepb_r – hzdept_r gives the horizon thickness.
ksat_r	µm/sec	chorizon	Saturated hydraulic conductivity, representative value (0.21–141.0 in this dataset). The primary infiltration rate coefficient and the single most important variable for MAR suitability. Normalized to [0,1] by dividing by 50 µm/sec. Contributes 50% of the infiltration sub-score.
awc_r	cm/cm	chorizon	Available water capacity, representative value (0.04–0.23). The volume of water the soil can hold between field capacity and wilting point. Contributes to the storage potential coefficient.
sandtotal_r	%	chorizon	Total sand content (5.5–95.0%). Higher sand content correlates with higher Ksat. Used as a cross-check on ksat_r and for texture-based Ksat estimation when ksat_r is null.

Column	Units	Source Table	Definition and QUBO Role
claytotal_r	%	chorizon	Total clay content (2.2–55.0%). Higher clay indicates clogging risk for MAR infrastructure. Used as a negative modifier on infiltration score and as a clogging risk indicator.
dbthirdbar_r	g/cm^3	chorizon	Bulk density at 1/3 bar (1.30–1.88). Used to calculate porosity as: porosity = 1 – (dbthirdbar_r / 2.65). Porosity determines the theoretical maximum storage volume.
om_r	%	chorizon	Organic matter content (0.03–13.0%). High organic matter indicates biological clogging risk for MAR infiltration basins. Used as a secondary penalty modifier.

3.4 Column Definitions — ssurgo_texture.csv

Column	Source Table	Definition and QUBO Role
cokey	chorizon	Component key. Join to ssurgo_soil_properties.csv on cokey.
chkey	chorizon	Horizon key. Together with cokey, uniquely identifies the soil layer this texture applies to.
texdesc	chtexturegrp	Texture group description. Human-readable texture name from the representative texture group.
texcl	chtexture	Texture class. USDA soil texture classification (14 classes in this dataset). Scored for QUBO: Sand=1.0, Loamy sand=0.9, Sandy loam=0.8, Loam=0.6, down to Clay=0.1. Contributes 20% of the infiltration sub-score.

The 14 texture classes present in the Georgia study area data, with their QUBO suitability scores:

Texture Class	Count	Score	MAR Suitability
Sandy clay loam	153	0.4	Low — high clay fraction impedes infiltration
Sandy loam	77	0.8	Good — balanced texture, adequate Ksat
Loamy sand	70	0.9	Very good — high sand content, high Ksat
Sandy clay	25	0.2	Poor — clay dominant despite sand presence
Sand	22	1.0	Excellent — maximum infiltration potential
Fine sandy loam	14	0.8	Good — similar to sandy loam
Loam	13	0.6	Moderate — balanced, adequate for some MAR
Clay	10	0.1	Very poor — essentially impermeable for MAR
Loamy coarse sand	5	0.9	Very good — coarse texture, high Ksat
Clay loam	4	0.3	Poor — clay-dominated
Silt loam	3	0.5	Moderate — fine particles reduce infiltration
Coarse sand	1	1.0	Excellent — maximum infiltration
Silty clay	1	0.15	Very poor — near-impermeable
Fine sand	1	1.0	Excellent — high infiltration

3.5 Column Definitions — ssurgo_water_table.csv

Column	Units	Definition and QUBO Role
mukey	—	Map unit key. Join to site_mukeys_{N}.csv. One row per mukey (aggregate, not per-horizon).
wtdepannmin	cm	Minimum annual depth to water table (0–153 cm where available). The shallowest the water table reaches in a typical year. Critical for exclusion: sites with wtdepannmin < 300 cm may have insufficient storage. NULL for many mukeys; NWIS groundwater data supplements this gap.
brockdepmin	cm	Minimum depth to bedrock. NULL for all mukeys in this dataset (typical for the deep sedimentary soils of the Dougherty Plain). Not used in Phase 1 QUBO.
niccdcd	code	National Irrigated Capability Class. Integer code (1–7) indicating irrigation suitability. Lower = more suitable. Potential secondary indicator for MAR infrastructure compatibility.

3.6 Column Definitions — site_mukeys_{N}.csv

Column	Units	Definition
site_id	—	Candidate site identifier (e.g., GA_001 through GA_050). Corresponds to the binary variable x_i in the QUBO.
latitude	decimal degrees	WGS84 latitude of the site centroid.
longitude	decimal degrees	WGS84 longitude of the site centroid.
mukey	—	SSURGO map unit key at this location. The join key to all three SSURGO tabular files.

3.7 Column Definitions — candidate_grid_{N}.csv

Contains the same site_id, latitude, and longitude columns as site_mukeys_{N}.csv but without the mukey column. This file records the grid geometry before spatial lookup. The site_mukeys file is the authoritative version for downstream processing.

4. Data Analysis

4.1 File: candidate_grid_{20,50}.csv

Structure: 3 columns (site_id, latitude, longitude), one row per candidate site.

Grid geometry: Regular grids within the bounding box [31.65°N–32.45°N, 84.55°W–83.65°W], with 0.05° inset from the bounding box edges. The 20-site grid has ~29.6 km × 21.2 km spacing (4 rows × 5 columns); the 50-site grid has ~14.8 km × 12.1 km spacing (7 × 8). The N=20 and N=50 grids share four corner coordinates by construction (both use numpy.linspace over the same bounding box with a 0.05° inset), so the union covers 66 unique candidate locations.

Downstream use: These coordinates are used by fetch_ssurgo_spatial.py (in scripts/ssurgo_scripts/) to query the SDA stored procedure for the mukey at each point. They also serve as the spatial reference for all other data sources (NWIS, NLCD, NOAA, OSM), which must be spatially assigned to the same grid. The grid coordinates feed directly into the pairwise distance calculation for M_{ij} (Equation 8 of the paper):

$$M_{ij} = \eta \cdot \exp(-d_{ij} / l)$$

where d_{ij} is the haversine distance computed from these lat/lon values, $\eta = 0.3$ is the penalty strength, and l = 10 km is the decay length.

4.2 File: site_mukeys_{20,50}.csv

Structure: 4 columns (site_id, latitude, longitude, mukey), one row per candidate site. All sites in the N=20 and N=50 grids have populated mukeys (zero nulls).

Unique mukeys: 20 in the 20-site grid and 40 in the 50-site grid. The two grids share corner coordinates and additional lattice points where indices align, so the 66-site union is served by approximately 50–60 unique mukeys depending on grid overlap with SSURGO polygon boundaries.

Downstream use: This is the primary join table connecting site locations to SSURGO soil properties. To build the QUBO coefficient for site i, join site_mukeys on mukey to ssurgo_soil_properties, ssurgo_texture, and ssurgo_water_table.

Transformation required: The mukey column should be cast to string type before joining, as it may be read as integer by pandas.

4.3 File: ssurgo_soil_properties.csv

Structure: 16 columns. Multiple rows per mukey (one per component × horizon combination). The column set covers 37 unique soil series names across the study area.

Key statistics: ksat_r ranges from 0.21 to 141.0 µm/sec (mean 30.3, median 9.0). The median of 9.0 indicates that most horizons in this region have moderate-to-low conductivity, consistent with the sandy clay loam and sandy loam textures that dominate the Dougherty Plain. Hydrologic soil groups are 49% B, 18% C, 14% B/D, 8% A, with the remainder split among A/D, C/D, and D.

Aggregation to site level: Each candidate site maps to one mukey, but each mukey may have multiple components and each component has multiple horizons. To produce a single infiltration score per site, the following aggregation is required:

1. Filter to the surface horizon (hzdept_r = 0) or the shallowest horizon for each component, as the surface layer controls initial infiltration rate.

2. For each component, extract ksat_r, hydgrp, and the texture class (joined from ssurgo_texture.csv on cokey + chkey).

3. Compute the component-level infiltration score, clamped to [0, 1]: $S_{comp} = 0.5 \cdot (ksat_r / 50.0) + 0.3 \cdot score_{hydgrp} + 0.2 \cdot score_{texture}$.

4. Weight-average across components using comppct_r: $S_i = \Sigma(comppct_r \cdot S_{comp}) / \Sigma\ comppct_r$.

5. Convert to minimization cost: $linear_dict[x_{site}] \mathrel{+}= 0.30 \cdot (1.0 - S_i)$.

Additional derived metrics: Porosity is calculated for each horizon and used in the recharge potential formula from Equation (13) of the paper:

$$porosity = 1 - (dbthirdbar_r / 2.65)$$

$$R_i = 100 \cdot (0.4 \cdot \tilde{K}_i + 0.3 \cdot \tilde{P}_i + 0.2 \cdot \tilde{T}_i + 0.1 \cdot (1 - \tilde{D}_i))$$

where $\tilde{P}_i$ is the normalized porosity, $\tilde{K}_i$ is normalized Ksat, $\tilde{T}_i$ is normalized texture score, and $\tilde{D}_i$ is normalized depth. Clay content (claytotal_r) and organic matter (om_r) contribute to a clogging risk penalty that may be included in the Phase 1 risk-aware scoring variant.

Machine learning considerations: No machine learning is required for the Phase 1 SSURGO-to-QUBO pipeline. The scoring functions are deterministic weighted combinations. However, for Phase 2 (real-data deployment with the OQI proposal), a supervised learning model could be trained on historical MAR performance data to learn optimal feature weights, replacing the manually set 0.5/0.3/0.2 split. Additionally, if ksat_r values are missing for some horizons, a pedotransfer function[14] (regression model trained on sand/clay/OM percentages) can estimate Ksat from the available texture data.

4.4 File: ssurgo_texture.csv

Structure: 4 columns. Joins to ssurgo_soil_properties.csv on cokey + chkey.

Content: 14 unique USDA texture classes. The most common is Sandy clay loam (153 rows, 38%), followed by Sandy loam (77, 19%) and Loamy sand (70, 18%). Only the texcl column (the USDA texture class) is used in the QUBO; texdesc is a human-readable equivalent.

Transformation: Map texcl to a numeric score using the lookup table defined in Section 3.4. This score feeds into the infiltration sub-score calculation described above. The mapping is applied as a simple dictionary lookup, no machine learning required.

4.5 File: ssurgo_water_table.csv

Structure: 4 columns, one row per mukey. Joins to site_mukeys on mukey.

Data completeness: wtdepannmin is populated for approximately 47% of mukeys; where available, values range from 0 to 153 cm. The remaining null values are typical for well-drained upland soils in the Dougherty Plain where the water table is deep enough that SSURGO surveys did not characterize it. brockdepmin is null for all mukeys, consistent with the deep sedimentary geology of the region (no shallow bedrock).

Transformation for QUBO: Sites where wtdepannmin is populated and less than 300 cm should receive an exclusion penalty (e_i approaches 1). Sites where wtdepannmin is null are assumed to have adequate depth (the absence of a water table record in SSURGO generally indicates deep water table). The NWIS groundwater monitoring data provides a complementary, time-series-based depth-to-water measurement that supplements the SSURGO static estimates — particularly for the mukeys where SSURGO data is null.

niccdcd transformation: The irrigation capability class (1–7) can optionally be used as a secondary suitability indicator. Lower values indicate better irrigation suitability, which correlates with MAR infrastructure compatibility. For Phase 1, this field is informational and not included in the core QUBO formulation.

4.6 Data Quality Notes

- A small number of records in ssurgo_soil_properties.csv have null values across all columns except cokey and mukey. These likely represent miscellaneous land components (e.g., water, pits) and should be filtered out during aggregation.
- The hydgrp column contains 7 values including dual groups (A/D, B/D, C/D). The dual designation indicates soils that behave differently when drained vs. undrained. For MAR, where infiltration basins add water, the undrained (second letter) rating should be used as the conservative estimate.
- ksat_r values vary by 3 orders of magnitude (0.21–141.0). The normalization divisor of 50 µm/sec (used in the VQ-MAR Data Specification) clips the top ~8% of values to 1.0. This is intentional — extremely high Ksat may indicate fractured rock or gravel channels that could transport contaminants, making the marginal benefit above 50 µm/sec negligible for MAR.
- The 50-site grid samples more of the soil landscape diversity than the 20-site grid; the union of the two captures the bulk of the soil heterogeneity in the study area.

5. References

Citations are listed in the order they appear in the body text. Inline superscript markers in the text refer to the corresponding entries below.

[1] Aseeri, S. A., Burgett, J., Melmer Stiefkens, J., Taylor, T., & Milewski, A. M., 2026, "VQ-MAR: A Variational Quantum-Classical Hybrid Framework for Managed Aquifer Recharge Site Selection," Submitted to IEEE International Conference on Quantum Computing and Engineering (QCE), Quantum End-to-End Hybrid Case Studies (QECS) Track for consideration.

[2] USDA Natural Resources Conservation Service (NRCS), Soil Survey Geographic Database (SSURGO). https://www.nrcs.usda.gov/resources/data-and-reports/soil-survey-geographic-database-ssurgo

[3] USDA NRCS Soil Data Access (SDA) web service: https://sdmdataaccess.nrcs.usda.gov. REST endpoint: https://sdmdataaccess.nrcs.usda.gov/Tabular/post.rest

[4] SDA_Get_Mukey_from_intersection_with_WktWgs84: NRCS stored procedure for point-in-polygon spatial lookup. Documentation: https://sdmdataaccess.nrcs.usda.gov/QueryHelp.aspx

[5] The Dougherty Plain is a karst physiographic district of the Gulf Coastal Plain in southwest Georgia, underlain by the Upper Floridan Aquifer system. Hicks, D. W., Gill, H. E., and Longsworth, S. A., 1987, Hydrogeology, chemical quality, and availability of ground water in the Upper Floridan Aquifer, Dougherty Plain, Georgia: USGS WRIR 87-4004.

[6] Managed Aquifer Recharge (MAR): Dillon, P. et al., 2019, Sixty years of global progress in managed aquifer recharge: Hydrogeology Journal, 27, 1–30.

[7] Qiskit: open-source SDK for quantum computing developed by IBM. https://qiskit.org. Packages: qiskit-optimization, qiskit-algorithms, qiskit-ibm-runtime.

[8] Hydrologic soil groups: USDA NRCS National Engineering Handbook, Part 630, Chapter 7. Group A = high infiltration (sand/gravel), B = moderate, C = slow, D = very slow (clay). Dual groups (A/D, B/D, C/D) indicate drained/undrained behavior.

[9] USGS National Water Information System (NWIS). Water Services REST API: https://waterservices.usgs.gov

[10] USGS National Land Cover Database (NLCD) 2021, MRLC Consortium: https://www.mrlc.gov/data/type/land-cover

[11] NOAA National Centers for Environmental Information (NCEI), Access Data Service v1: https://www.ncei.noaa.gov/access/services/data/v1

[12] OpenStreetMap contributors. https://www.openstreetmap.org. Queried via Overpass API: https://overpass-api.de

[13] Quantum Approximate Optimization Algorithm (QAOA): Farhi, E., Goldstone, J., and Gutmann, S., 2014, A Quantum Approximate Optimization Algorithm: arXiv:1411.4028.

[14] Pedotransfer functions: empirical regression models that estimate soil hydraulic properties (e.g., Ksat) from more readily available soil properties (sand/clay/OM content). See: Schaap, M. G. et al., 2001, ROSETTA: a computer program for estimating soil hydraulic parameters: J. Hydrology, 251, 163–176.